

Power-Aware Test Optimization for Core-Based 3D-SOCs under TSV-Constraints

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Abstract—While 3D chips open up versatile potentialities in compact system design, they pose the challenge of testing the composite system, which consists of multiple cores, logic, and memory, interconnected across different layers of the chip. The test strategy for such chips must also take into account the issues of inherent power and thermal constraints, design of test-access mechanism (TAM), and the decision concerning pre-bond and post-bond test choices. Additionally, for post-bond testing, the constraints imposed by the limited use of TSVs, worsen the controllability and observability of the cores that are accessed through the inter-layer scan-paths. Thus, while designing the TAM architecture, the optimization of overall test time under the constraints of power and TSV-count, is needed. This paper presents a new technique for test-time reduction in post-bond core-based 3D-SOCs, considering certain constraints on test power and TAM width (i.e., bounds on TSVs). The proposed algorithm runs much faster compared to prior art, and our results on several ITC02 benchmarks reveal significant reduction in test-time for most of the cases.

Keywords: SOC, TAM, Through-silicon-via (TSV).

I. INTRODUCTION

Recent advances of 3D vertical integration on silicon now offer immense potential in fabricating heterogeneous and complex systems as a single package. Such a composite system may consist of a set of embedded digital, analog, and mixed-signal cores interconnected as a single device. These cores may represent a variety of functional blocks such as processors, memories, DSP units, application specific hardware blocks, memories, network controllers, or peripherals. With the advancement of technology, more and more number of cores can be assembled as a single system-on-chip (SOC) [1].

Concomitant to the increasing design-complexity of 3D chips, the test-complexity continues to grow rapidly as evident from the increasing need for test data volume as well as test time. Minimization of test time had been studied in the past as a natural objective function while addressing test-problems in these SOC. Additionally, test-power needs to be reduced as much as possible in order to mitigate thermal damage that may be caused while exercising tests. Power consumption during testing often causes overheating of the chip and may render the chip partially or fully unusable. Hence, the test vectors needed for testing various cores on the SOC are to be designed and ordered in such a way that the power-constraint is satisfied while minimizing the overall test time [2]. A special access method called test-access-mechanism (TAM) and core wrappers were introduced to manage the test session

in complex SOC. The TAM wires are used to transport test data from source to the core-under-test and to collect test responses from the core to the test-response sink [3]. The core wrapper, as its name suggests, provides an interface between a core and the TAM. Note that a 3D-TAM will be needed to transport test data from various layers of the 3D-SOC to the bottom layer. Through-silicon-vias (TSVs) are generally used in implementing 3D-TAMs. TSVs are inter-layer vertical vias that are used for interconnecting nets among different layers or dies [4]. They are needed mainly for signal-routing, power/ground routing, thermal sinking, and for transmitting clock signals. Since all the pins and sinks are embedded at the bottom layer in a 3D-IC, TSVs also play a major role in the design of 3D-TAMs. However, they occupy considerable area on a chip, introduce IR-drop at each coupling point, and are quite costly. Therefore, the test-access architecture should be designed in such a way that the number of TSVs used for this purpose is reduced as far as possible.

Note that the power consumption of a circuit during test mode often turns out to be much higher than that needed in the functional mode. While concurrent testing of multiple cores may reduce the test time significantly, it might generate a high peak temperature in the circuitry, which enhances the risk of thermal damage. Because of the inherent limitations of TAM, test-concurrency issues are highly dependent on how the set of cores is partitioned into TAM groups, which in turn, depends on how the cores are partitioned and placed into different layers. Hence, thermal-aware placement of cores is also essential in order to ensure the convenience of testing [5].

In this paper, we address the problem of minimizing the overall test time of a digital-SOC by placing the cores in suitable TAM partitions. Note that the cores, which are placed in the same partition are to be tested serially. In order to facilitate heat dissipation, we adopt the strategy of placing the modules that are more thermally active, towards the bottom-layers. This ensures that the temperature profile among the TAM partitions will follow a non-increasing gradient from bottom-to-top layers. Consequently, fewer thermal-TSVs will be required to dissipate heat during testing. Some additional constraints such as an upper-bound on TSV-count, the placement of cores in different TAM partitions are also considered. In our analysis, we ignore the overhead for pre-bond test cost and time.

The rest of this paper is organized as follows: Section II presents related prior work considering various TAM optimizations. In Section III, we present a simple example to illustrate the TAM architecture that we have used. Section IV gives the

problem formulation and Section V describes our proposed algorithm. In Section VI, we present experimental results on ITC-02 SOC benchmarks [1]. Finally, Section VII summarizes the conclusions and future work.

II. PRIOR WORK

In the past, test time optimization for 3D-ICs was performed by using Integer Linear Program (ILP) and mixed-ILP (MILP) solvers [6], [7]. However, ILP techniques are computationally expensive, and they might require substantial amount of CPU-time while solving the problem for large SOC. Heuristic algorithms formulated on the basis of greedy approaches may perform much better in terms of CPU-time as compared to exact methods such as ILP. Pradhan et al. [8] suggested a greedy heuristic to solve this problem; however, in their work, only two-layer chips were considered and power management issues were ignored. They used bin packing [9] method to generate the initial partitions and then tried to optimize it by a cost-based heuristic. Power-constrained test scheduling, where a test power consumption threshold is specified for the SOC, has later been widely studied in the literature [7], [10]. Unfortunately, not much work has been reported in case of 3D-SOCs. In the work by Wu et al. [7], no power or thermal constraints were reported. Our proposed algorithm runs much faster and produces better results for most of the test cases compared to previous work [7], [8] in terms of test time, test-power, and TSV-count.

III. AN EXAMPLE OF THE ARCHITECTURE

In Fig. 1, in one TAM partition two cores (M_1 and M_2) are placed in Layer 1 (the bottom layer), in another TAM partition two other cores (M_3 and M_4) are placed in Layer 2 and finally one core (M_5) is placed in Layer 3 (the top layer). Let W be the total width of the TAM available for testing that are used to access the cores. As shown in Fig. 1, the total number of wires that will be required under the TAM partition (TAM_1) consisting of cores M_1 and M_2 will be T_1 , for another TAM partition (TAM_2) consisting of cores M_3 and M_4 will require T_2 and finally for the last TAM partition (TAM_3) consisting of core M_5 will require T_3 number of wires. Since, neither TAM_2 nor TAM_3 is the bottommost layer, for these cores we require 3D- TAMs called as test elevators [11]. For the test elevators between layer 1 and layer 2, we require $2(T_2 + T_3)$ number of TSVs, as $(T_2 + T_3)$ for test data in and the rest for test data out. Similarly, between layer 2 and layer 3 we require $2T_3$ number of TSVs. Hence, for testing purpose only, we require exactly $2T_2 + 4T_3$ number of TSVs in the suggested SOC and so the TSV constraint should be $\geq 2T_2 + 4T_3$. As $W = T_1 + T_2 + T_3$, the total number of test pins is $2(T_1 + T_2 + T_3)$. Now, in this work, we have followed a strategy of placing the TAM partitions in such a way that the TAM partition of highest width is used in the bottommost layer and the rest of the TAM partitions are placed above that layer in decreasing order of their width, which ensures that the total number of TSVs required will be less. We also make an effort to place the TAM partitions with high power consuming cores towards the bottom. So, we would expect the bottommost TAM partition to have the highest width as well as consume maximum power and thereby become the most heated TAM partition also. Finally to ensure minimum testing time as well as better heat sinking, it is important to assign the cores to a certain TAM partition in an efficient manner, however, we also have to make sure that it satisfies the TSV-constraint.

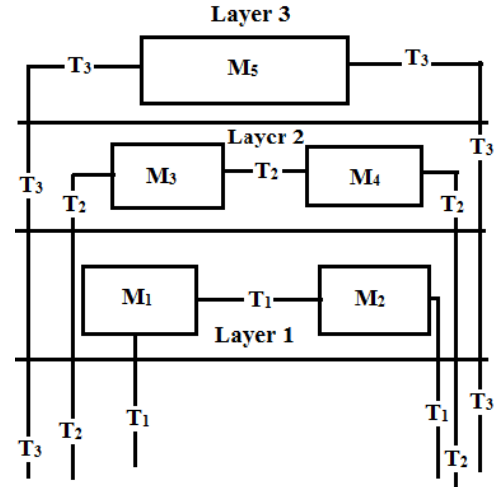


Fig. 1: Example of TAM usage with TSV-constraint

IV. PROBLEM FORMULATION

The objective of our algorithm is to minimize the time needed for testing the SOC while satisfying the constraints imposed on TAM width, thermal bound, and the number of TSVs. The multi-layer TAM partitioning problem can be naturally modeled as a graph multi-partitioning problem. As usual, cores are denoted by vertices (V) and their TAM interconnects are denoted by edges (N).

An SOC is typically modeled as a graph $G = (V, N)$ with a set of cores V and a set of TAM wires N , where $V_i \in V$ denotes V_i is a core in the graph G . We have to partition V in such a way that $\bigcup_{j=1}^K Tam_j = V$ and $\bigcap_{j=1}^K Tam_j = \emptyset$, where K is the number of TAM partitions. Let $Time_{Tam_j}$ be the total test time required to test the TAM partition j . Then our objective is to minimize $(\max_j (Time_{Tam_j}))$ as much as possible so that we can test the SOC in minimum time.

A. Cost Definition

Let us assume that initially the cores are arbitrarily partitioned into several mutually exclusive partitions and core V_i is placed in Tam_j . We define $Cost(V_i, Tam_j) = \alpha * \max_j (Time_{Tam_j}) + (1 - \alpha) * Layer(Tam_j) * Pwr(V_i)$, where $Layer(Tam_j)$ is the layer where the Tam_j is placed and $Pwr(V_i)$ is the power consumed during testing of that particular core. Note that the cost is defined as a convex combination of test time required and power consumed by a core controlled by the parameter $\alpha \in [0, 1]$. Basically if $\alpha = 1$, then we are optimizing only on the basis of test time and if $\alpha = 0$, then only on the basis of power consumption. Also note that the value of power consumption is scaled by the layer number and hence if a higher layer consumes more power, it affects the cost more than a lower layer consuming the same amount of power. For quantifying the benefit of moving a core V_i to a new partition Tam_j from its previous partition Tam_{prev} , we define $Gain(V_i, Tam_j) = Cost(V_i, Tam_{prev}) - Cost(V_i, Tam_j)$. So for each core a separate gain will be defined for each of the TAM partitions. Finally, we define the global gain of each pass as $Gain(G) = \max_{i,j} (Gain(V_i, Tam_j))$. In each step we move only that module (say V_{max}) which will give the maximum positive gain; ties will be broken arbitrarily by choosing one of the cores with maximum positive

gain. After moving the core V_{max} , we will again evaluate $Cost(V_i, Tam_j)$ and repeat the whole step again till in a certain step no core can be found which gives a positive gain ($Gain(G)$). Incidentally, we allow the same core to change partition more than once and even to come back to a partition which it left earlier as we found out that it sometimes leads to better optimization. Since we proceed if and only if there is a positive gain, the method is guaranteed to converge. In Subsection V-A, we illustrate the implications of the above cost definition through a typical example.

V. PROPOSED WORK AND ALGORITHM

We already mentioned that we place the TAM partitions from bottom to top layer in order of non-increasing width. The rationale behind this is that the TAM partitions with smaller width when placed in upper layers will demand less number of TSVs to build the test elevator. Our algorithm finds an efficient assignment of the cores among multiple TAM partitions such that the overall test time is minimized, also it ensures that the most power consuming modules are placed in the TAM partition that is mapped to the bottommost layer. The runtime of our algorithm is negligible compared to that of previous works [7], [8]. Our technique comprises two major phases, as described in Algorithm 1 and Algorithm 2. In Algorithm 1, we have performed an initial TAM partitioning following the 2D rectangular bin packing method [12], where the bin is having a width of W (TAM width) and length T (test time) and our objective is to place the bins in such a way that it would require as less area as possible. Then in Algorithm 2, on the basis of the initial TAM partitions, we calculate the Gain for each core when placed in each of the TAM partitions. The for-loop starting at step 3 is used to map the different TAM partitions into layers from bottom to top based on the descending order of their width as obtained from Algorithm 1. The for-loop starting in Step 7 finds the cost of each core for each TAM partition and the one starting in Step 14 finds the core with overall maximum gain. Finally, we move the core with the maximum gain from its current TAM partition to the TAM partition, where it has obtained maximum gain. This iterative process is continued until no further optimization can be done.

input : A set of cores V , each core V_i with a set of K test times under K TAM partitions ($Time(V_i, Tam_1), \dots, Time(V_i, Tam_K)$), the total TAM width (W) and width of K TAM partitions ($W_1, \dots, W_K$)

output: Initial assignment of each core to a certain TAM partition

```

1 for each core  $V_i$  do
2    $min \leftarrow W_1 * Time(V_i, Tam_1)$ ;
3    $P_i \leftarrow 1$ ; /*  $P_i$  is the TAM partition,
   where core  $V_i$  is placed */
4   for  $j \leftarrow 2$  to  $K$  do
5     if ( $min > (W_j * Time(V_i, Tam_j))$ ) then
6        $min \leftarrow W_j * Time(V_i, Tam_j)$ ;
7        $P_i \leftarrow j$ ;
8     end
9   end
10 end

```

Algorithm 1: Initial Multi-Partitioning in 3D-SOC

input : A set of cores V , each core V_i with a set of K test times under K TAM partitions ($Time(V_i, Tam_1), \dots, Time(V_i, Tam_K)$), the TAM partition P_i where core V_i is initially assigned, the total TAM width (W) and width of K TAM partitions ($W_1, \dots, W_K$)

output: Near optimum assignment of the cores maintaining TSV and thermal constraint

```

1 Sort TAM partitions in descending order
  according to there TAM width;
2 If TAM widths are same then sort them in
  descending order of power consumption of its
  initial assigned cores;
3 for  $j \leftarrow 1$  to  $K$  do
4    $Layer(Tam_j) \leftarrow j$ ;
5 end
6  $max \leftarrow 0$ ;
7 for each core  $V_i$  do
8   for  $j \leftarrow 1$  to  $K$  do
9     calculate  $Cost(V_i, Tam_j)$ ;
10    if ( $P_i = j$ ) then
11       $Cost(V_i, Tam_{prev}) \leftarrow$ 
         $Cost(V_i, Tam_j)$ ;
12    end
13  end
14  for  $j \leftarrow 1$  to  $K$  do
15    calculate  $Gain(V_i, Tam_j)$ ;
16    if ( $max < Gain(V_i, Tam_j)$ ) then
17       $max \leftarrow Gain(V_i, Tam_j)$ ;
18       $CoreToMove \leftarrow i$ ;
19       $Tam_{Move} \leftarrow j$ ;
20    end
21  end
22 end
23 if ( $max > 0$ ) then
24   Place module  $V_{CoreToMove}$  to  $Tam_{Move}$ ;
25   update power dissipation, test time of TAMs;
26   go to step 6;
27 end
28 STOP;

```

Algorithm 2: Convex Multi-Partitioning in 3D-SOC

A. A Typical Example

Let us consider an example as shown in Table I and Table II. Table I gives the test times for each of the cores when placed in the three given TAM partitions having TAM widths of 3, 5 and 7 respectively. The cores with numbers 1 and 4 are placed initially in the TAM partition having the width of 3, core 5 and 6 are placed in the partition of width 5 and the rest are placed in that of width 7. The assigned partition for each core as per Algorithm 1 is reflected in Table I, by showing the corresponding test times in bold fonts. Let us now analyze the test times for this placement, say for example, if core 3 is placed in the TAM partition of width 7, according to the bin packing method it would acquire an area of $7 * 2 = 14$ square units, whereas if placed in the other TAM partition of width 5(3), it requires $5 * 4 = 20$ ($3 * 6 = 18$) square units. Hence 14 being the least, core 3 is placed in the TAM partition of width 7. So, in this way each core goes to a certain TAM partition. After the initial placement of cores, in this example, the total test time turns out to be $\max(22 + 10, 4 + 5, 5 +$

TABLE I: TAM width, test time, *Cost*, *Gain* of a typical example

TAM	Test time						α	<i>Cost</i>						<i>Gain</i>					
	1	2	3	4	5	6		1	2	3	4	5	6	1	2	3	4	5	6
3	22	15	6	10	7	9	0.7	29.6	55.4	40.1	31.4	32.7	39.5	0	-25.5	-13.2	0	-6.7	9.9
5	13.5	10	4	7	4	5		20.55	37.4	31.4	21.4	26	29.6	9.05	-7.5	-4.5	10	0.0	0.0
7	10	5	2	5	3	4		14.3	29.9	26.9	18.4	24.2	26	15.3	0	0	13	1.8	3.6
3	22	15	6	10	7	9	0.3	26.4	66.6	42.9	30.6	24.3	37.5	0	-39.5	-22.8	0.0	-6.3	-11.1
5	13.5	10	4	7	4	5		17.95	44.6	30.6	20.6	18	26.4	8.45	-17.5	-10.5	10	0.0	0.0
7	10	5	2	5	3	4		10.7	27.1	20.1	13.6	13.8	18	15.7	0.0	0.0	17.0	4.2	8.4

3) = 32. In Table II, we have provided the power dissipation of each core during testing. Then for different values of α , using the expressions from Subsection IV-A, for each core V_i , we have calculated the $Cost(V_i, Tam_j)$ for the initial placement scheme as well as when V_i is placed in the other TAM partitions. Then we have calculated $Gain(V_i, Tam_j)$. In Table I, we have shown the results for two different values of α ; $\alpha = 0.7$ designating more emphasis on test time reduction and $\alpha = 0.3$ for priority on power constraint. For example when $\alpha = 0.7$, $Cost(1, Tam_3) = (0.7 * 32 + 0.3 * 3 * 8) = 29.6$, $Gain(1, Tam_7) = (0.7 * (22 + 10) + 0.3 * 3 * 8) - (0.7 * (10 + 5 + 2) + 0.3 * 1 * 8) = 29.6 - 14.3 = 15.3$. From Table I, we find that the highest positive gain occurs if core 1 moves to the TAM partition of width 7 (shown in bold fonts) and the 2nd highest gain happens for core 4 when placed in that same partition. The calculation in this case being $Gain(4, Tam_7) = (0.7 * 32 + 0.3 * 3 * 10) - (0.7 * 22 + 0.3 * 1 * 10) = 31.4 - 18.4 = 13$. So we place core 1 in the TAM partition of width 7, at the end of 1st pass. Then the test time will be $\max(10, 4 + 5, 10 + 5 + 2) = 17$ whereas the power value at the lowest layer will change from 40 units to 48 units.

However, when $\alpha = 0.3$, the maximum positive gain occurs for core 4, which shows the dependence on the parameter α , even in this small example. Table III reflects the power dissipation values after a single pass of the Algorithm 2. In

TABLE II: Power dissipation values of the 6 cores

Power dissipation of cores						
Cores	1	2	3	4	5	6
Power dissipation	8	25	15	10	6	12

the second pass also, we perform similar computations and find that $Gain(4, Tam_5) = (0.7 * 17 + 0.3 * 3 * 10) - (0.7 * 17 + 0.3 * 2 * 10) = 20.9 - 17.9 = 3$. So, core 4 now has the highest gain and it should be placed in the TAM partition of width 5. Note that after the 1st pass, seeing the second best gain we may have been tempted to place core 4 in the TAM partition of width 7, however as core 1 moved after the first pass, the whole basis of gain calculation changed, showing the efficacy of multiple passes. We stop the above procedure when in a certain pass the highest gain turns out to be non-positive. In this small example it stopped after 2 (4) passes for $\alpha = 0.7$ ($\alpha = 0.3$).

TABLE III: Power dissipation at various layers

TAM width	At Layer	After Algorithm 1	After 1st pass of Algorithm 2	
			$\alpha = 0.7$	$\alpha = 0.3$
7	1	40	48	50
5	2	18	18	8
3	3	18	10	18

B. Complexity Analysis

Let $|V|$ be the number of cores and K be the number of TAM partitions in the 3D SOC. In Algorithm 1, for each of the cores, we have done $O(K)$ computation. Hence the total time required is $O(K * |V|)$.

In Algorithm 2, steps 1 and 2 run in $O(K \log K)$ time for sorting the K partitions. The algorithm needs to visit each core once for each of the TAM partitions to calculate the Cost and the Gain, hence Step 6 onward takes $O(|V| * K)$ time. Hence a single pass of Algorithm 2 runs in exactly $O(|V| * K + K \log K)$ time, which will be $O(K * |V|)$ as $|V|$ will obviously dominate over $\log_2 K$. Now the overall time complexity depends on how many times the goto statement in Step 26 gets executed. Note that the number of passes would have been exactly bounded by $|V|$, if like some conventional algorithms, we had put a lock on each core after it moves once. In that case, the worst-case time complexity would have been $O(K * |V|^2)$. We intentionally did not stop repeated movements of a core to keep open the chance for further optimization. However, it does not cause any harm in efficiency, as in practice the number of passes comes out to be approximately around $|V|$, as shown in Table IV.

VI. EXPERIMENTAL RESULTS

Our partitioning algorithm is implemented in C (Linux environment) and we have tested it on an Intel CoreI3 machine with a primary memory of 1GB. Note that we have calculated test times for the individual cores in the same way as described by Wu et al. [7]. The power dissipation for each core has been calculated using the data available from a weblink [13].

The results were obtained by running the algorithm on the ITC02 benchmarks [1] and CPU times were also noted. The execution time required was negligible even for the larger benchmarks and is far less compared to that reported in earlier works. For each SOC, we have compared our experimental results with that of Wu et al. [7], and to make the comparisons justified, we have considered exactly the same TAM partitions that they had used in their experiments. Manjari et al. [8] had experimented only with bi-partitions and their algorithm did not address thermal issues, hence we did not show their results in the tables here. However, for thermal oblivious cases, we have made comparisons with their results elsewhere [14].

Extensive experiments were conducted for different TAM widths and different multi-partitions for all the SOC's by varying the convex parameter α from 0 to 1 at an interval of 0.05. In Table IV, we report all aspects of our experiment viz. TSV-count, test time, CPU Time and layer-wise power dissipation values for the largest benchmark p93791 with 32 cores. For brevity, we present them only for two different values of α , 0.35 and 0.80. All the parameters when compared with the work of Wu et al. clearly demonstrate the efficacy of our proposed algorithm. The test times and the number of TSVs used were better than them in almost all cases and as

TABLE IV: Exp. Results of multi-partitioning on p93791 (32 cores)

$\alpha = 0.35$												
Layer	TAM Partition	TSV-Limit		3D Test Time		CPU time		No. of Passes	Power unit consumption			
		Ours	[7]	Ours	[7]	Ours	[7]		Layer1		Layer2	
2	3, 13	6	80	1911161	2103700	0.0	8.2	9	11351.52		3026.88	
	4, 20	8	80	1408153	1406320	0.0	22.0	9	11351.52		3026.88	
	5, 27	10	80	1008607	1405440	0.0	160.6	9	11998.91		2379.49	
	5, 35	10	80	992751	1241245	0.00	160.6	11	12004.49		2373.91	
	5, 43	10	80	888792	1405440	0.0	160.6	9	11999.77		2378.63	
	5, 51	10	80	573720	1152364	0.0	160.6	11	13919.13		459.27	
	5, 59	10	80	573720	1008220	0.0	160.6	11	13919.13		459.27	
3									Layer1	Layer2	Layer3	
	1, 1, 14	6	80	2005814	2229152	0.01	1.9	13	12106.10	1894.39	377.90	
	1, 4, 19	12	80	1412113	1412113	0.00	3.5	20	13523.72	839.50	15.18	
	1, 7, 24	18	80	946604	1009262	0.00	6.9	18	7979.0	6380.78	18.62	
	1, 16, 23	36	80	734000	999931	0.00	53.8	26	5853.65	8184.47	340.28	
	7, 16, 25	60	80	641170	995650	0.00	96.3	10	4769.35	9103.81	505.23	
	13, 19, 24	90	80	561893	995522	0.00	302.8	9	6070.66	7088.46	1219.27	
4, 25, 35	66	80	575005	995522	0.00	46.6	17	4907.93	9124.77	345.70		
4									Layer1	Layer2	Layer3	Layer4
	1, 1, 2, 12	14	80	1946902	4023112	0.02	2.2	22	12973.79	1284.29	13.12	107.19
	1, 1, 1, 21	12	80	1503815	3823747	0.01	7.0	14	13378.07	351.22	0.86	648.25
	1, 1, 4, 26	18	80	1023723	3695454	0.03	2.0	17	12997.63	1037.45	343.32	0.00
	1, 10, 1, 28	30	80	846151	2699795	0.01	55.4	22	7429.93	6932.43	15.18	0.86
	4, 13, 1, 30	48	80	731269	1903700	0.03	220.5	35	6269.04	6454.94	1651.22	3.20
	4, 1, 25, 26	72	80	557484	1121150	0.03	252.2	28	6070.66	7587.97	706.01	13.76
1, 10, 25 28	96	80	509566	995522	0.05	339.7	39	4527.49	8743.96	1101.21	5.73	
$\alpha = 0.80$												
Layer	TAM Partition	TSV-Limit		3D Test Time		CPU time		No. of Passes	Power unit consumption			
		Ours	[7]	Ours	[7]	Ours	[7]		Layer1		Layer2	
2	3, 13	6	80	1911161	2103700	0.0	8.2	9	11351.52		3026.88	
	4, 20	8	80	1408153	1406320	0.0	22.0	9	11351.52		3026.88	
	5, 27	10	80	1008607	1405440	0.0	160.6	9	11998.91		2379.49	
	5, 35	10	80	992751	1241245	0.00	160.6	11	12004.49		2373.91	
	5, 43	10	80	888792	1405440	0.0	160.6	9	11999.77		2378.63	
	5, 51	10	80	573720	1152364	0.0	160.6	11	13919.13		459.27	
	5, 59	10	80	573720	1008220	0.0	160.6	11	13919.13		459.27	
3									Layer1	Layer2	Layer3	
	1, 1, 14	6	80	2005814	2229152	0.01	1.9	13	12106.10	1894.39	377.90	
	1, 4, 19	12	80	1410725	1412113	0.00	3.5	16	12087.74	2268.85	21.81	
	1, 7, 24	18	80	946604	1009262	0.00	6.9	15	7979.0	6372.88	26.55	
	1, 16, 23	36	80	742629	999931	0.00	53.8	18	5856.80	8499.79	21.81	
	7, 16, 25	60	80	641170	995650	0.00	96.3	8	4769.35	9097.43	511.61	
	13, 19, 24	90	80	561893	995522	0.00	302.8	9	6070.66	7079.76	1227.97	
4, 25, 35	66	80	575005	995522	0.00	46.6	17	4907.93	9118.39	352.08		
4									Layer1	Layer2	Layer3	Layer4
	1, 1, 2, 12	14	80	1946445	4023112	0.02	2.2	17	13523.72	724.81	22.67	107.19
	1, 1, 1, 21	12	80	1503815	3823747	0.01	7.0	14	13378.07	351.22	648.25	0.86
	1, 1, 4, 26	18	80	1023723	3695454	0.03	2.0	17	12997.63	1029.55	351.22	0.00
	1, 10, 1, 28	30	80	846151	2699795	0.01	55.4	16	7429.93	6921.09	26.52	0.86
	4, 13, 1, 30	48	80	715686	1903700	0.02	220.5	18	7054.31	6767.43	534.85	21.81
	4, 1, 25, 26	72	80	563763	1121150	0.03	252.2	28	6080.44	7903.31	380.19	14.45
1, 10, 25 28	96	80	502275	995522	0.05	339.7	29	5747.11	7479.13	1136.91	15.25	

already mentioned the CPU times were less by several orders. We could not compare the power values of individual layers as they did not report any. However, in the rightmost multi-column of this table, we give an idea of how in some cases, our method has been able to bring down power dissipation in the upper layers, which is very much intended. In Fig. 2, Fig. 3 and Fig. 4, we present the results for the other three benchmarks for $\alpha = 0.35$, which as per our observation gives the most balanced results. For brevity, in these cases we have omitted the report on power values and CPU times, for most of the cases our CPU times are negligible (0.0) where as their CPU times vary from 1.9 - 225.7, 0.2 - 407, 0.8 - 46.8 for the three benchmarks p34392, p22810 and d695 respectively. For p34392, again our results were found to be better in almost all aspects. For d695 and p22810, our test times were sometimes worse but with respect to TSV count, we achieved much better results.

Overall, for all the benchmarks, our run-times are always much better, whereas the test times are less compared to prior work in most of the cases. The most interesting fact is that our algorithm reported improved test time while using a significantly fewer number of TSVs especially for larger benchmarks, and hence, it thoroughly reduces the other overheads associated

with them.

VII. CONCLUSION AND FUTURE WORK

We have proposed a fast multi-partitioning algorithm for TAM architecture optimization for 3D-SOCs, where the minimization of test time is achieved while maintaining the constraints on TSV-count and test power dissipation. Experimental results reveal that our method is quite effective in reducing test time. Additionally, the proposed algorithm will help in reducing the overall production cost of the chip as the total TSV-count is observed to decrease significantly. As a future task, we look forward to improve the partitioning algorithm in order to take care of other constraints such as area and delay.

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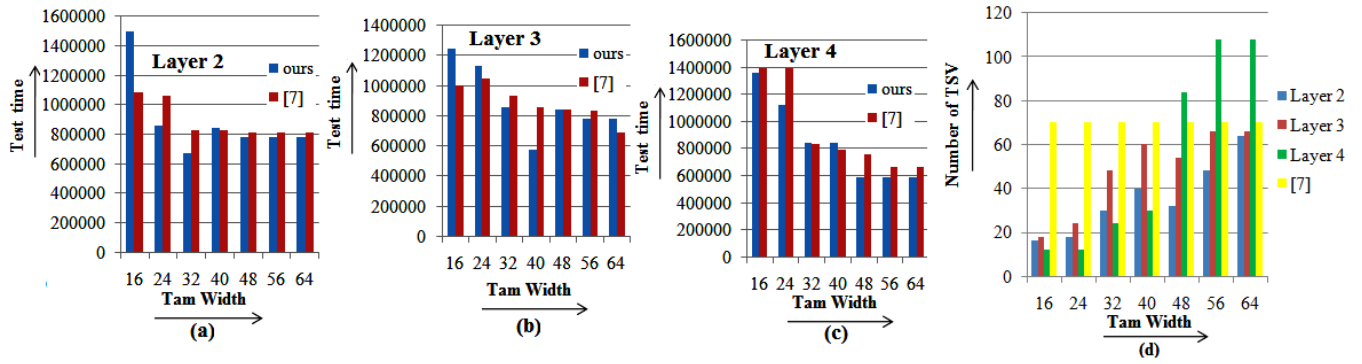


Fig. 2: Variation of test-time and TSV-count with TAM width for p34392

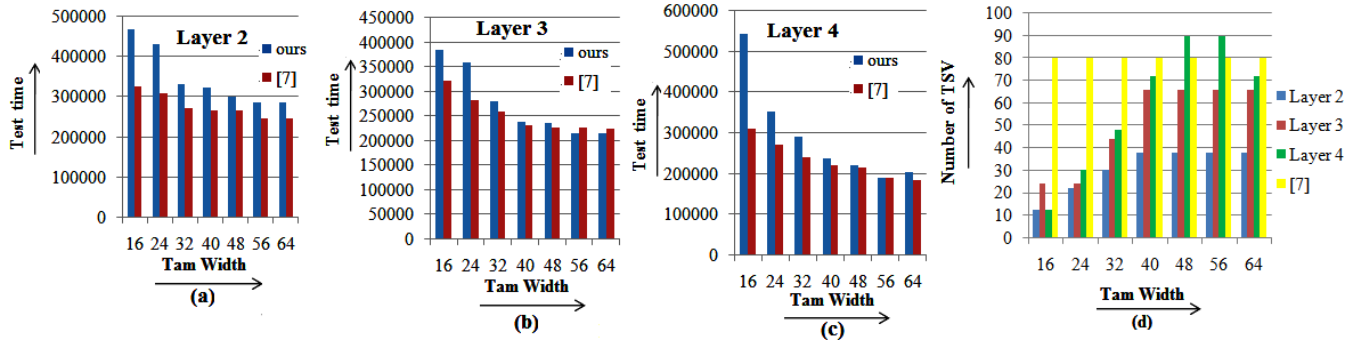


Fig. 3: Variation of test-time and TSV-count with TAM width for p22810

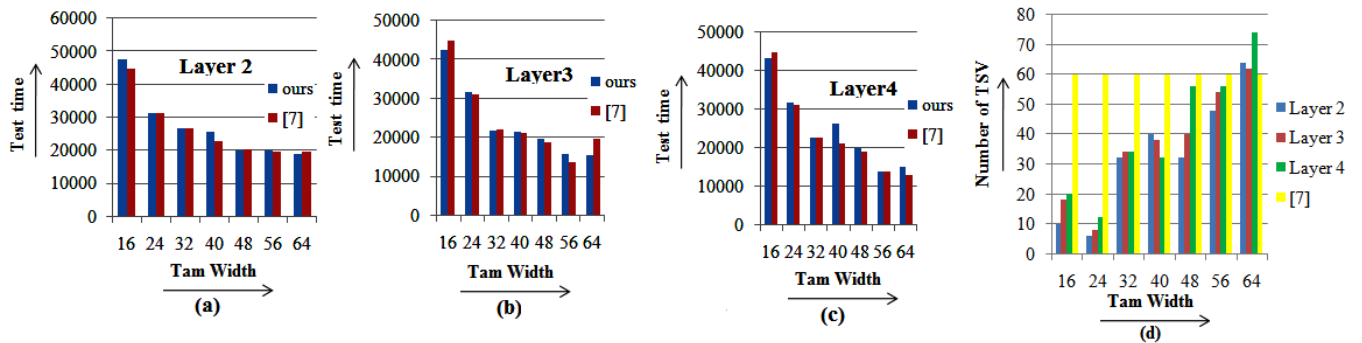


Fig. 4: Variation of test-time and TSV-count with TAM width for d695

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